

# BLUESTOCK FINANCIAL ANALYTICS PLATFORM

## ANALYST USER GUIDE & REST API MANUAL

*A comprehensive technical guide for financial analysts, data engineers, and quantitative developers.*

|                    |                                                         |
|--------------------|---------------------------------------------------------|
| Document Version:  | 1.0.0 (Production Release)                              |
| Target Audience:   | Financial Analysts, Data Engineers, Equity Researchers  |
| Platform Coverage: | Nifty 100 Universe (92 Companies, 10 Relational Tables) |
| Publication Date:  | September 2026                                          |

# 1. System Architecture & Platform Infrastructure

The BlueStock Financial Platform is engineered as a multi-tier analytics ecosystem combining high-throughput data ingestion, automated data quality validation, quantitative ratio calculations, interactive Streamlit web dashboards, and high-concurrency FastAPI REST endpoints.

## Key Components:

- **SQLite Database Engine (nifty100.db)**: 10 relational schema tables enforcing primary key uniqueness, foreign key constraints, and index optimizations.
- **ETL Normalisation Pipeline**: Cleans ticker symbols and standardizes financial year strings across source files.
- **FastAPI REST Server (port 8000)**: Serves 16 endpoints for integration with web applications, external BI tools, and data pipelines.

# 2. Automated Data Quality (DQ) Validation Engine

The ETL pipeline automatically executes 16 Data Quality rules (DQ-01 to DQ-16) prior to database loading. Any CRITICAL failure halts loading to preserve data integrity, while WARNING logs are written to output/validation\_failures.csv.

| Rule ID | Rule Name             | Target Table     | Severity | Description                                             |
|---------|-----------------------|------------------|----------|---------------------------------------------------------|
| DQ-01   | PK Uniqueness         | Companies        | CRITICAL | Validates company_id and ticker uniqueness.             |
| DQ-02   | FK Integrity          | All Child Tables | CRITICAL | Enforces foreign key relationships with companies.      |
| DQ-03   | Balance Sheet Balance | Balancesheet     | WARNING  | Verifies Assets = Total Liabilities + Equity.           |
| DQ-04   | OPM Cross-Check       | Profitandloss    | WARNING  | Cross-checks reported OPM vs computed operating margin. |

### 3. Financial Ratio Engine & CAGR Edge Cases

The ratio engine computes 50+ financial KPIs per company-year across profitability, leverage, cash flow quality, and multi-year CAGR compounding.

#### Edge Case Handling:

- **Negative Equity Handling:** ROE returns *None* if equity capital + reserves  $\leq 0$ .
- **Banking Sector Carve-Out:** D/E high leverage flags are suppressed for Financials companies.
- **Interest Coverage Infinity:** Debt-free companies with 0 interest expense are assigned  $\$ICR = \infty$ .

### 4. Interactive Stock Screener Engine

The stock screener allows analysts to filter the 92-company universe using 10 custom threshold sliders or 6 preset strategies.

#### Preset Screening Strategies:

1. **Quality Compounder:** ROE > 16.5%, D/E < 0.35, FCF > 0, 5Y Rev CAGR > 10%.
2. **Value Pick:** P/E < 22, P/B < 3.5, D/E < 0.8, Dividend Yield > 1.0%.
3. **Growth Accelerator:** PAT CAGR 5Y > 12%, Revenue CAGR 5Y > 10%, D/E < 0.8.
4. **Dividend Champion:** Dividend Yield > 1.5%, Payout < 70%, FCF > 0.

## 5. Peer Percentile Rankings & Radar Visualizations

Percentile rankings (PERCENT\_RANK) are computed across 10 metrics within each of the 11 peer groups. Debt-to-Equity is inversely ranked ( $1 - \text{PERCENT\_RANK}$ ) so lower leverage ranks higher.

### 8-Axis Polar Radar Charts:

Each company in a peer group features a PNG polar radar chart in reports/radar\_charts/ displaying the company's filled polygon overlayed against the sector peer group average.

## 6. Valuation Engine & Overvaluation Flags

The valuation engine calculates Free Cash Flow (FCF) Yield % and compares company P/E multiples against sector median P/E to assign overvaluation flags:

- **Caution (Overvalued):**  $P/E > 1.5 \times \text{Sector Median P/E}$
- **Discount (Undervalued):**  $P/E < 0.7 \times \text{Sector Median P/E}$
- **Fair:** All other company valuations.

## 7. Streamlit Web Dashboard Navigation (8 Screens)

Launch the web dashboard via streamlit run app.py to access 8 interactive screens:

1. **01\_home.py**: Top 6 summary KPI tiles, sector breakdown donut chart, top 5 quality companies table.
2. **02\_profile.py**: Autocomplete search, 6 KPI cards, 10Y Revenue/PAT bar chart, ROE/ROCE line chart, Pros & Cons badges.
3. **03\_screener.py**: 10 metric sliders, 6 preset buttons, live filtered table, CSV export button.
4. **04\_peers.py**: 11 peer groups dropdown, Plotly polar radar chart, side-by-side KPI comparison table.
5. **05\_trends.py**: Multi-metric overlay line chart over 10 years with YoY % change annotations.
6. **06\_sectors.py**: Sector dropdown, bubble scatter chart (Revenue vs ROE vs Market Cap), sector median bar chart.
7. **07\_capital.py**: Plotly Treemap of 92 companies grouped by 8 capital allocation patterns.
8. **08\_reports.py**: Company annual report search with BSE PDF links and fallback badges.

## 8. PDF Report Generation & Publishing

The platform uses ReportLab to batch render 2-page company tearsheet PDFs, 11 sector PDF reports, and a 92-page portfolio summary PDF.

### Report Locations:

- Company Tearsheets: reports/tearsheets/\_tearsheet.pdf (92 PDFs)
- Sector Reports: reports/sector/\_report.pdf (11 PDFs)
- Portfolio Summary PDF: reports/portfolio/portfolio\_summary.pdf

## 9. REST API Developer Guide (FastAPI)

Start the REST API server via `uvicorn src.api.main:app --port 8000`. All endpoints are hosted under prefix `/api/v1`.

### Example API Calls:

```
curl http://localhost:8000/api/v1/health
curl http://localhost:8000/api/v1/companies/TCS
curl http://localhost:8000/api/v1/screener?min_roe=15&max_de=1.0
curl http://localhost:8000/api/v1/sectors/IT/companies
```

## 10. Troubleshooting & System Maintenance

Troubleshooting common issues during operation:

- **Port Conflict (8000 or 8501 occupied):** Kill running background process or launch uvicorn/streamlit on alternative ports.
- **Database Missing Error:** Run make load to re-populate nifty100.db.
- **Unit Test Failures:** Run make test to execute 80+ unit tests across all 7 packages.

*End of Analyst User Guide — BlueStock Financial Analytics Platform (v1.0.0)*

# Appendix 1: Technical Specifications & Database Schema

## DDL

Complete DDL definitions for companies, profitandloss, balancesheet, cashflow, financial\_ratios, peer\_percentiles, and sectors tables.